CS 498: HW1 Report

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1 Rigid fitting and RANSAC

Table 1: Rigid registration

	SVD	**RANSAC**
Translation error (mm)	4.71	4.52
Rotation error (deg)	0.0433	0.0338
Inlier ratio	0.655[a]	0.650[b]
Inlier RMSE (mm)	17.28[a]	13.38[b]

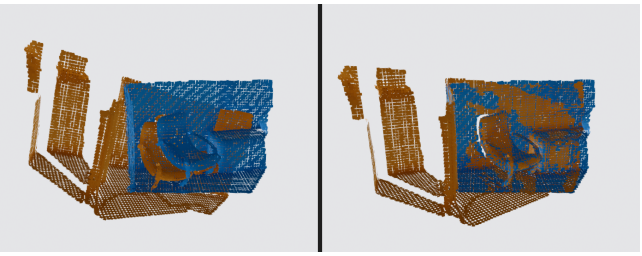
The threshold in the RANSAC is used to decide how close a pair must be to regard as agreeing with the current hypothesis. If it is too small, true pairs might be rejected because of noises, leading to unreliable counts of inlier across different hypotheses. If it is too big, the outliers might be accidentally accepted, making them bias the least-squares fit, which is what RANSAC is designed to prevent.

2 Real-data ICP and odometry

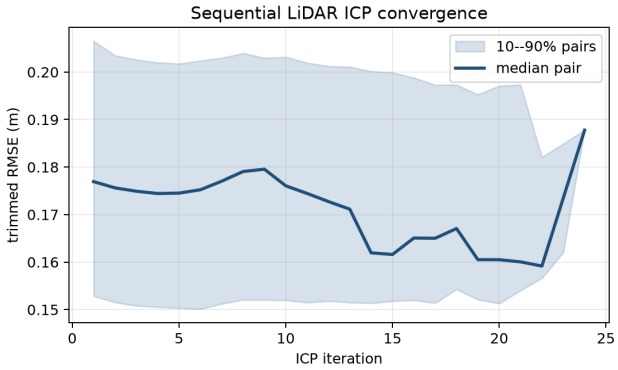
All the output figures are in the Figure 1, and the video showing Open3D demo pair for the chair is `icp_iterations.mp4`, output along with the video `icp_odometry_online.mp4`. In the online odometry video, there are two panels shown side by side of BEV centered on the LiDAR. The left panel displays the current sweep of every point, which are the local features that we can use for alignment. The right panel is accumulating intensity results from all sweeps up to the current frame. On the right panel, there is a incrementing cyan trajectory overlay showing the past and current vehicle poses. Good local alignment would keep curbs and facades sharp and relatively static across the different frames. These good alignments would be updated on time and reflected correspondingly in the current frame from the latest sweep. Drift seems to be blurry and duplicated with noticeable offset between the old accumulated view and the current sweep. The trajectory is mentioned above, specifically, and it draws out a progressing path composed by the past predicted vehicle poses.

Demo final RMSE: 0.0155377 m; Planar XY odometry ATE: 5.49867 m; mean one-step translation RTE: 0.0217497 m; mean one-step rotation RTE: 0.122454.

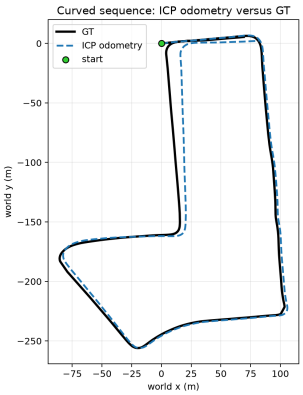
As we see the Figure 1d, we can see the first big drift occurs on the bottom left of the image, and the mapped structure there is much more blurry and less sharper compared with other structures. On the ICP odometry, the path at the end going from the bottom to the top also reveals double edges on the right walls, which also explains big drift in the this part of trajectory.



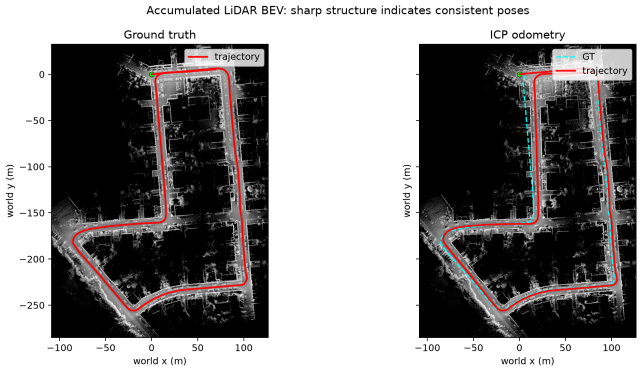
(a) Open3D demo pair before and after ICP



(b) ICP convergence (RMSE per iteration)



(c) GT and ICP 2D trajectories



(d) GT and ICP accumulated BEV maps

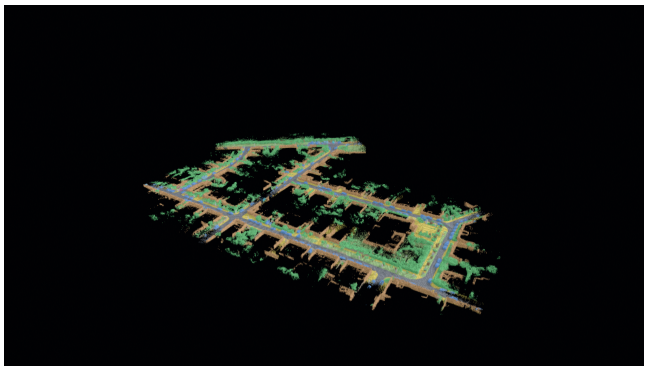
Figure 1: Task 3: ICP alignment and LiDAR odometry results

3 Semantic voxel mapping

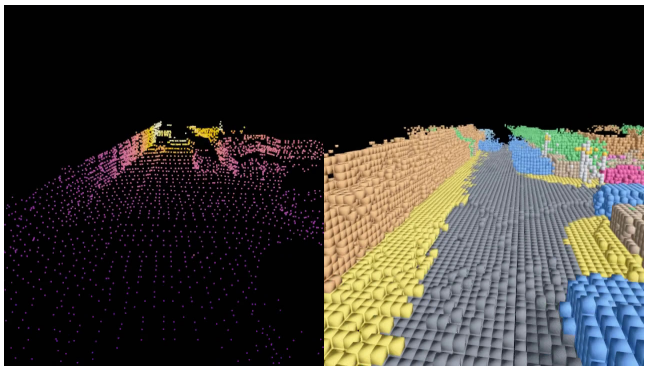
The grid has shape $1063 \times 1355 \times 38$ at 0.25 m, which is 54,733,870 voxels. Storing a float32 score for 19 classes needs $54{,}733{,}870 \times 19 \times 4$ bytes ≈ 4.16 GB. Since LiDAR only measures the surfaces, so the rest grid is empty and the space sensor never saw. This explains why the map could be sparse. Hence, we can only store occupied voxels which has the number of 843,990 and the $843{,}990 \times 19 \times 4$ bytes $\approx$ 64 MB storing data.

Table 2: Semantic voxel mapping results.

Metric	Value
Frames aggregated	1,024
Occupied voxels	843,990
Voxel accuracy	1.000
Mean IoU	1.000
Per-class IoU (17 present classes)	1.000 (all)
Mean observations per voxel	3.03
Maximum observations per voxel	36
Mean normalized entropy (mean log evidence)	0.0273



(a) Open3D shoulder-view semantic voxel map



(b) Side-by-side LiDAR/map follow video

Figure 2: Task 4: semantic voxel map

The file names are `semantic_lidar_follow.mp4` and `semantic_occupancy.mp4`. In the json file, `voxel_coordinates_correct` is true. On the 14 second of the video, we can see that a blue parked car sit against the yellow sidewalk and the gray road. So a single voxel can contain points from the car, the curb, and the road, which is competing. Also, a green tree canopy hangs over the brown building facade. Weighting every observation equally neglects the decaying performance of LiDAR on the distant objects. For example, far points are often noisier and they should have lower confidence comapred with nearer points. So it is easier to result in competing evidence with the use of far points. Smaller voxels can preserve very fine structures like curbs and clear boundaries. However, it occupies much more data size. With larger voxels, even though the data memory is fewer, we would also lose tiny structures and get blurry points.

4 Failure analysis

The most noticeable visual failure is still the problem we have described in Figure 1d, the ghosted walls result in big deviation in the odometry result. This is because of our assumption in the registration: each LiDAR scan is only aligned to a single previous scan. Relative transform errors make chained poses compound more errors. If we can use average out the transforms by using more scan pairs, we might be able to get better results.